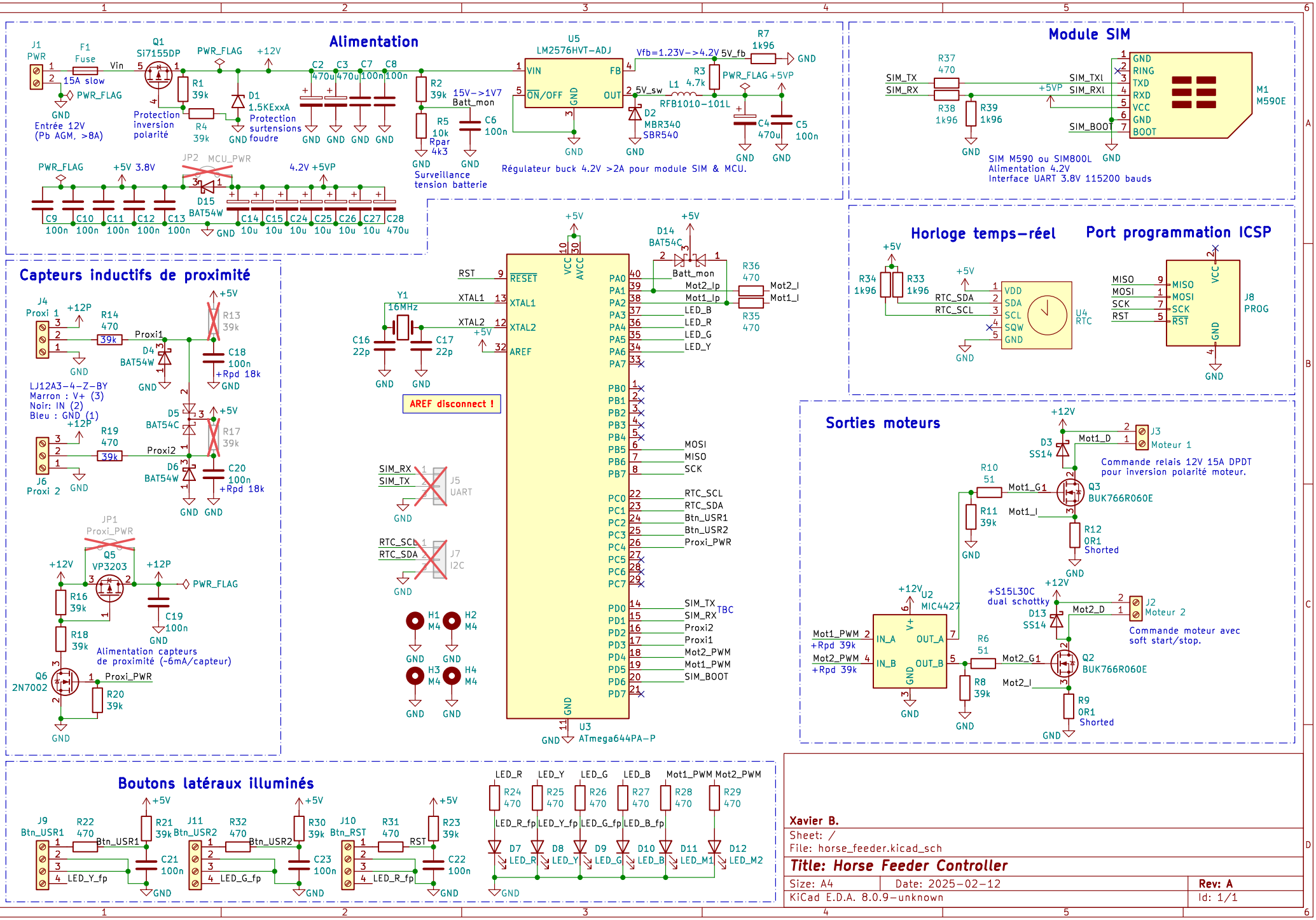
Alimentation
J1 PWR
F1 Fuse
15A slow
Entrée 12V (Pb AGM, >8A)
Q1 SI7155DP
PWR_FLAG
Protection inversion polarité
D1 1.5KExxA
Protection surtensions foudre
C2 C3 C7 C8 470u 470u 100n 100n
U5 LM2576HVT-ADJ
VIN
ON/OFF
FB
OUT
GND
Vfb=1.23V->4.2V
5V_fb
R7 1k96
R2 39k
15V->1V7
Batt_mon
R5 10k Rpar 4k3
C6 100n
Surveillance tension batterie
5V_sw
L1 4.7k
RFB1010-101L
D2 MBR340 SBR540
R3
+5VP
C4 470u
C5 100n
Régulateur buck 4.2V >2A pour module SIM & MCU.
PWR_FLAG
+5V 3.8V
JP2 MCU_PWR
D15 BAT54W
4.2V +5VP
C9 C10 C11 C12 C13 100n
C14 C15 C24 C25 C26 C27 C28 10u
C29 470u
Module SIM
R37 470
R38 1k96
R39 1k96
SIM_TX
SIM_RX
SIM_TXI
SIM_RXI
SIM_BOOT
M1 M590E
SIM M590 ou SIM800L
Alimentation 4.2V
Interface UART 3.8V 115200 bauds
Horloge temps-réel
R34 1k96
R33 1k96
RTC_SDA
RTC_SCL
U4 RTC
Port programmation ICSP
J8 PROG
MISO
MOSI
SCK
RST
Capteurs inductifs de proximité
J4 Proxi 1
J6 Proxi 2
+12P
R14 470
R19 470
39k
Proxi1
Proxi2
D4 BAT54W
D5 BAT54C
D6 BAT54W
R13 39k
R17 39k
C18 100n
C20 100n
+Rpd 18k
LJ12A3-4-Z-BY
Marron : V+ (3)
Noir: IN (2)
Bleu : GND (1)
JP1 Proxi_PWR
Q5 VP3203
R16 39k
R18 39k
C19 100n
Alimentation capteurs de proximité (~6mA/capteur)
Q6 2N7002
R20 39k
Proxi_PWR
Y1 16MHz
C16 22p
C17 22p
AREF disconnect !
SIM_RX
SIM_TX
J5 UART
RTC_SCL
RTC_SDA
J7 I2C
H1 H2 H3 H4 M4
U3 ATmega644PA-P
RESET
XTAL1
XTAL2
AREF
VCC
AVCC
PA0 Mot2_Ip
PA1 Mot1_Ip
PA2 LED_B
PA3 LED_R
PA4 LED_G
PA5 LED_Y
PB5 MOSI
PB6 MISO
PB7 SCK
PC0 RTC_SCL
PC1 RTC_SDA
PC2 Btn_USR1
PC3 Btn_USR2
PC4 Proxi_PWR
PD0 SIM_TX
PD1 SIM_RX
PD2 Proxi2
PD3 Proxi1
PD4 Mot2_PWM
PD5 Mot1_PWM
PD6 SIM_BOOT
D14 BAT54C
Batt_mon
R36 470
R35 470
Mot2_I
Mot1_I
Sorties moteurs
J3 Moteur 1
Mot1_D
D3 SS14
Commande relais 12V 15A DPDT pour inversion polarité moteur.
R10 51
R11 39k
R12 0R1 Shorted
Mot1_G1
Mot1_I
Q3 BUK766R060E
U2 MIC4427
+S15L30C dual schottky
J2 Moteur 2
Mot2_D
D13 SS14
Commande moteur avec soft start/stop.
Mot1_PWM
Mot2_PWM
+Rpd 39k
R6 51
R8 39k
R9 0R1 Shorted
Mot2_G1
Mot2_I
Q2 BUK766R060E
Boutons latéraux illuminés
J9 Btn_USR1
J11 Btn_USR2
J10 Btn_RST
R22 470
R32 470
R31 470
R21 39k
R30 39k
R23 39k
C21 100n
C23 100n
C22 100n
LED_Y_fp
LED_G_fp
LED_R_fp
R24 470
R25 470
R26 470
R27 470
R28 470
R29 470
LED_R_fp
LED_Y_fp
LED_G_fp
LED_B_fp
D7 LED_R
D8 LED_Y
D9 LED_G
D10 LED_B
D11 LED_M1
D12 LED_M2
Xavier B.
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Title: Horse Feeder Controller
Size: A4
Date: 2025-02-12
Rev: A
KiCad E.D.A. 8.0.9-unknown
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